Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A Signal and Image Processing

Exercise 3

Fourier Series and Fourier Transform

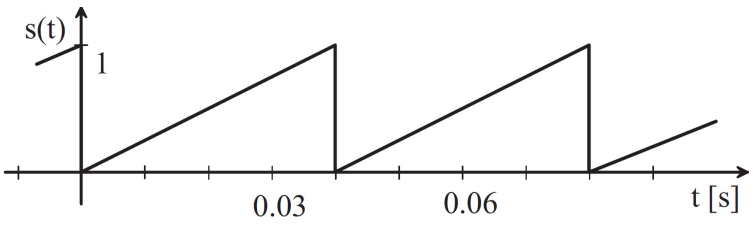
3.1. A periodic signal can be described with a Fourier series.

3.1.1. Write the definition of the Fourier series and how the complex Fourier coefficients are calculated.

3.1.2. What does the Fourier series say about the composition of a periodic signal?

3.1.3. Consider the signal $s(t)$ in the figure below. What period T_0, basic frequency f_0, and angular frequency ω_0 does the signal $s(t)$ have?

3.1.4. Compute the Fourier components of the signal given below.



3.2. The signal $s(t)$ in 3.1.5 passes through an ideal low-pass filter with cut-off frequency $f_c = 60$ Hz. What frequencies are there in the output signal $y(t)$?

3.3. Decide the amplitude spectrum for the continuous-time signal $x(t)$ in the figure below.

3.4. Decide the Fourier transform of the signal $x(t) = e^{-at}\sin(\omega_0 t)\,u(t)$. You can use a table.

3.5. Assume the system H to be linear.

3.5.1. What defines a linear system?

3.5.2. What defines a time-invariant system?

3.5.3. What defines a causal system? What is specific with its impulse response? Give an example of a difference equation of a non-causal discrete-time system.

3.6. The input to a LTI system contains the frequencies 5 Hz, 125 Hz and 1000 Hz. What are the frequency components of the output? Could there be fewer frequencies in the output?

3.7. Write the convolution sum for a general discrete-time system.

3.7.1. Write the convolution sum for a causal discrete-time system.

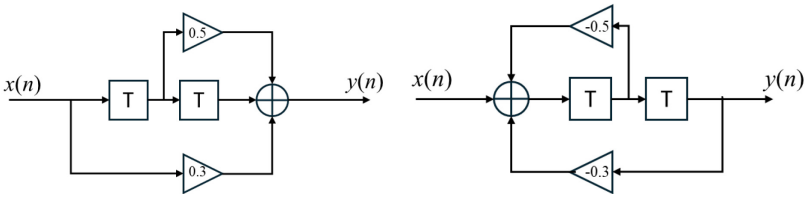
3.7.2. Write the convolution sum for a general discrete-time system with a causal input signal.

3.8. Consider the discrete-time systems below.

3.8.1. Write the difference equations.

3.8.2. Compute the impulse response by iterating for $n = 0, 1, 2, 3, 4$

3.8.3. What is such a system called (considering the length of the impulse response).



3.9. A linear time-invariant, and causal discrete-time system is defined by its impulse response $h(n) = e^{0.5n} \sin(0.146n)\, u(n)$.
Compute the output $y(n)$ for $n = 0, 1, 2, 3, 4$ by convolving the impulse response with the input signal
$x(n) = \{x(0), x(1), x(2), x(3)\} = \{1, 2, 1, -1\}$

3.10. Draw the circuit diagram for a general IIR filter with M forward connections and N feedback connections.

3.10.1. Set the requirements on the parameters to assure it to be an IIR filter.

3.10.2. Set the requirements on the parameters to assure it to be an FIR filter.

3.11. What is the consequence of having a limited number of samples when carrying out a Fourier analysis?

3.11.1. Describe mathematically in both time domain and frequency domain the consequence of using $N < \infty$ samples of the signal $s(n)$, $0 \leq n < \infty$. What happens in both time and frequency domains when N is increased?

3.12. Search for four different windows (for windowing) on the Internet. Give their respective mathematical expressions. Describe the size of main lob and sidebands for these.

3.12.1. Which of these windows would you choose to identify two neighbouring frequencies of about the same amplitudes?

3.12.2. Which of these windows would you choose to identify two neighbouring frequencies with rather different amplitudes?

3.13. Derive the requirement on the impulse response $h(n)$ for a discrete-time system to be BIBO stable.

3.13.1. Let the impulse response of a discrete-time system be $h(n) = a^n$. Set the restriction on a for the system to be BIBO stable.

3.14. A signal is sampled in four points, and one gets the sample values 1, -1, 0 and 1. Compute the DFT.